Hood Wheel Ratio Test (power version: no encoders yet)

Which hood wheel power makes the ball spin the least?

Team: _____ Date: _____ Names: _____

The big idea

Our shooter squeezes the ball between **two wheels**: a **big flywheel** (about 4 inches) in front and a **small hood wheel** (about 2 inches) in the back. If the two wheel surfaces move at *different* speeds, the ball starts to **spin**. A spinning ball can curve or bounce in a way we do not want.

The small wheel has to turn **faster** to keep up with the big wheel. The number that tells us how much faster is the **ratio**:

Ratio = hood wheel power ÷ flywheel power

Example: flywheel 30% power and ratio 2.00 → hood wheel = 30 × 2.00 = **60% power**

Our guess: a ratio near **2.00** gives the least spin. **Our job:** test it, and find out if the best ratio stays the same when the flywheel goes faster or slower.

> **For now we set POWER (a percent)** because the encoders are not connected yet. Power is not exactly the same as speed: it changes with the battery. Use a **fresh battery** and write down the volts. A wheel cannot go over **100%**.

SAFETY — everybody must follow these

- ☐ **Eye protection on** for everyone near the shooter.
- ☐ **Nobody stands in front** of the shooter or in the path of the ball.
- ☐ **Hands, hair, sleeves and lanyards stay away** from the wheels.
- ☐ **Wheels keep spinning after you stop them.** Wait until they are completely stopped before you reach in.
- ☐ Anyone can yell **"STOP!"** The Driver presses **X** right away.
- ☐ **Only a coach clears a jam.**

Jobs (one person can do two jobs if you are a small group)

Driver	Holds controller 1. Turns the dials, starts and stops the shooter, flips the gate. Says "Shooting!" before each shot.
Feeder	Holds controller 2. Runs the intake (Right Bumper) to load balls.
Video	Films every shot in **slow motion**. Says the shot number out loud.
Recorder	Writes everything in the data table. Reads the screen.
Spotter	Watches where the ball lands and watches for safety. Calls "STOP!" if needed.

Set up (do this once)

- ☐ Fresh battery. Battery volts: _____
- ☐ **Coach check:** the gate positions are done. Coach initials: _____
- ☐ Put a tape mark on the floor. Shooting distance: _____ inches. **Use the same mark for every shot.**
- ☐ Draw a **thick marker stripe** all the way around the ball. This is how we see the spin.
- ☐ Set the phone to **slow motion** on a steady stand where it can see the ball leave the shooter.
- ☐ On the Driver Station choose **"Shooter Ratio - Alpha"**. Press INIT. The gate should say **LOADING**. Press START.

Controls

Controller 1 (Driver)

B / A	Flywheel power up / down (5%)
D-pad Right / Left	Ratio up / down (0.05)
Y	Shooter ON
X	**Shooter OFF** (STOP)
Left Trigger	Flip the gate

Controller 2 (Feeder)

Right Bumper	Intake: pull balls in
Left Bumper	Reverse the intake

Reading the screen

Wheels settled? YES = OK to shoot. **WAIT** = the 3 second count is not done.
CAPPED = the hood wheel cannot go over 100% power. **Skip that row.**

For every row of the data table

1. **Driver:** set the flywheel power (B / A) and the ratio (D-pad) to match the row. Check the hood wheel power on the screen matches the table. If it says CAPPED, skip the row.
2. **Driver:** press **Y**. Wait for **Wheels settled? YES**.
3. **Feeder:** load a ball. **Driver:** say "Shooting!" and flip the gate with the **Left Trigger**. Flip it back after the ball leaves.
4. **Video** says "Shot 1". **Recorder** fills in the spin code and the landing letter (see below).
5. Wait for **Wheels settled? YES** again. (The count starts over every time the gate flips.) Repeat until you have **5 shots**.
6. **Recorder:** write the **battery volts** from the top of the Driver Station screen.
7. Go to the next row. (Changing a dial also starts the count over.)

How to write down a shot

Watch the slow-motion video. Follow the marker stripe from the moment the ball leaves until it lands.

Top box: spin code (a letter + a number)

B	**Backspin**: the top of the ball turns **back toward the robot**
N	**No spin**: the stripe hardly moves (use **N0**)
T	**Topspin**: the top of the ball turns **forward toward the target**

Number: 0 = none 1 = a little (stripe goes around less than 2 times) 2 = medium (2 to 6 times) 3 = a lot (more than 6). Examples: **B2**, **N0**, **T1**.

Bottom box: where it landed

S	**Short**: landed before the target
O	**On** the target
L	**Long**: landed past the target

> The wheels keep spinning after you press X. **Never reach in until they stop.**

Part 1 — Data Table (first distance)

Distance: _____ Date: _____ Recorder: _____

Each shot box: **top** = spin code (like B2) | **bottom** = landing (S, O or L). Hood power = flywheel power × ratio.

Flywheel power %	Ratio	Hood power %	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Battery volts	Notes
25	**1.80**	45							
25	**2.00**	50							
25	**2.20**	55							
25	**2.40**	60							
30	**1.80**	54							
30	**2.00**	60							
30	**2.20**	66							
30	**2.40**	72							
40	**1.80**	72							
40	**2.00**	80							
40	**2.20**	88							
40	**2.40**	96							
50	**2.00**	100							

Bonus row (last row): ask your coach first. The hood wheel is at 100% power, its limit. If the screen says CAPPED, do not record it. Every other row stays under 100%, so none of them should say CAPPED.

Part 2 — Change the distance, then the results

Part 2: second distance

Move the shooter to a **new distance** and mark it. Ask your coach which 3 flywheel powers reach the target from there. Use your **best ratio from Part 1**, and also the ratio **0.20 lower** and **0.20 higher** than it. Keep every hood wheel power at **100% or less**.

New distance: _____ Best ratio from Part 1: _____ Date: _____

Flywheel power %	Ratio	Hood power %	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Battery volts	Notes

Results

Flywheel power %	Ratio with the **least spin** (distance 1)	Spin code there	Ratio with the least spin (distance 2)
25			
30			
40			

Think about it

1. Was the best ratio about the **same** at every power? Circle one: **YES** **ALMOST** **NO**

2. Look at your **battery volts** column. Did the volts change a lot while you tested? Could that change the results?

3. What do you think is the reason for the best ratio you found?

4. Did the best ratio change when you changed the **distance**?